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MAIL SHIELD

SPAM VS HAM EMAIL CLASSIFIER

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PROJECT OVERVIEW

An AI-powered email security system that classifies emails into:

- 🟢 Ham (Normal / Promotional Email)
- 🔴 Spam (Suspicious Email)

Built using:

- Python
- Flask
- Machine Learning (TF-IDF + Logistic Regression)
- NLP + Rule-based security layers

INTRODUCTION & PROBLEM

1. Introduction

Spam vs Ham Email Classifier is a smart cybersecurity web application designed to analyze email content and detect whether it is safe or malicious.

It uses a **hybrid AI approach** combining:

- Machine Learning predictions
- Natural Language Processing (NLP)
- Rule-based security analysis

This ensures higher accuracy and better explainability than traditional spam filters.

1.1 Problem Statement

Email is one of the most common attack vectors in cybercrime.

Major issues:

- ❌ Phishing emails bypass normal filters
- ❌ Users cannot identify fake emails easily
- ❌ Existing systems are only binary (Spam / Not Spam)
- ❌ No explanation behind detection

Result: Users click malicious links → data theft, hacking, fraud

1.2 Solution

This project solves the problem by introducing:

- Multi-layer detection system
- Explainable AI scoring
- Real-time classification (< 1 second)
- API-based integration for systems

SYSTEM DESIGN & WORKING

2. System Architecture

The system uses **5 intelligent layers**:

Detection Layers

- Source Analysis (sender trust)
- NLP Text Analysis (keywords, urgency detection)
- Link Analysis (malicious URLs)
- Brand Checker (trusted vs fake brands)
- Machine Learning Model (TF-IDF + Logistic Regression)

Final Decision System

Each layer produces a score:

- Safe indicators reduce risk
- Suspicious indicators increase risk

Final Decision:

- Score ≤ 2 → Safe Email
- Score 2–5 → Promotional Email
- Score > 5 → Spam Email

ML Model Details

- Algorithm: Logistic Regression

- Features: TF-IDF (8000 features)
- Dataset: 10,000+ emails
- Accuracy: ~96%
- Training Split: 80/20

API Feature

- Endpoint: `POST /api/predict`
- Input: Email text
- Output: Label + Risk Score + Explanation

TARGET MARKET & SALES STRATEGY

3. Target Clients

Client Type	Why They Need It
Small Businesses	No cybersecurity team
Universities	High email traffic
Banks & Finance	High phishing risk
Freelancers	Handle sensitive data

4. How We Will Sell the Product

We use a **Freemium SaaS Model**:

Pricing Strategy:

- 🆓 Free Plan → 100 emails/day
- 💼 Pro Plan → $29/month unlimited use
- 👥 Team Plan → $99/month dashboard access
- 🏢 Enterprise → Custom deployment

Marketing Strategy

- Website landing page
- Social media ads (LinkedIn, Facebook)
- Direct outreach to IT departments
- Integration into email systems
- SaaS marketplaces (AWS / Azure)

5. Why Clients Will Buy It

Clients choose this product because:

Real-time detection (fast response)
Multi-layer AI security (hard to bypass)
Explainable results (transparent scoring)
Easy API integration
Low cost, high accuracy (96%+)
Works better than traditional spam filters

PROMOTION & FINAL DELIVERABLES

6. Project Poster (Design Idea)

Poster Content:

Headline:
Spam vs Ham Email Classifier

Subheading:
AI Powered Email Security System

Features Section:

- Real-time Detection
- 96% Accuracy
- Multi-Layer AI System
- Phishing Protection

Footer:
"Protect your inbox before it's too late!"

7. Promotional Video (Less than 5 Minutes)

Video Script:

1. Intro (0:00–0:30)

- Problem of phishing emails
- Cybercrime statistics

2. Product Introduction (0:30–1:30)

- Spam vs Ham Email Classifier
- What it does

3. How It Works (1:30–3:00)

- 5-layer system explanation
- ML + NLP demo

4. Benefits (3:00–4:00)

- Accuracy
- Speed
- API integration

5. Sales Pitch (4:00–5:00)

- Pricing model
- Target clients
- Call to action

FINAL SUMMARY

Spam vs Ham Email Classifier is a production-ready AI-based cybersecurity system designed to protect users from phishing attacks using:

- Machine Learning
- NLP
- Rule-based security layers
- REST API integration

It is:
Scalable
Accurate
Explainable
Commercially viable
